INCH-POUND

MIL-C-70461D (AR)
AMENDMENT 3
15 February 1994
SUPERSEDING
AMENDMENT 2
09 June 1993

MILITARY SPECIFICATION

COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS
FOR CHARGE, PROPELLING, 155MM, M203A1

This Amendment forms a part of Military Specification MIL-C-70461D (AR) dated 28 May 1991, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

PAGE 4

3.4 TABLE I: Delete all the entries under the heading of tensile strength and substitute with the following new entries:

	"Tensile Strength
Rear Cap	1614 psi (Minimum)
Igniter Container	812 lbs. (Minimum Peak Load)
Body, Molded	2600 psi (Minimum)"

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4.4.2.1 Major 108: Substitute "4.5.4" with "4.5.4.1".

PAGE 11

4.4.2.2 Add the following:

"Major 118 Tensile Strength 4.4.3.6 3.4 4.5.4.2"

PAGE 13

4.4.2.3 Add the following:

"Major 114 Tensile Strength 4.4.3.6 3.4 4.5.4.3"

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4.4.3.6 Delete in its entirety and substitute the following new paragraph:

"4.4.3.6 Tensile strength. Five (5) bodies, five (5) igniter containers and five (5) rear caps shall be randomly selected from the first article and each product lot of its respective component to provide the required test specimens for tensile tests in accordance with 4.5.4. Failure of any specimen shall be cause for lot rejection of its respective component."

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* 4.5.3.3.2.2 Delete in its entirety and substitute the following new paragraph:

"4.5.3.3.2.2 Diphenylamine (DPA) - UV Method. Cut a sample into 1/4 inch pieces or smaller. Dry to constant weight using a microwave moisture analyzer. Cool the sample to room temperature in a desiccator and weigh 1 gram specimen to the nearest 0.2 mg. Transfer the specimen into a beaker, add 50 ml of 90% aqueous methanol, and set the beaker in a warm ultrasonic bath for 10 minutes. Disperse the specimen in the liquid by means of a homogenizer for about 2 minutes or until the specimen is completely disintegrated. Decant the liquid portion into a filtering apparatus fitted with a suitable filter paper, and retain as much of the solid in the beaker. Add a similar amount of 90% aqueous methanol into the beaker and redisperse the specimen in the liquid by means of the homogenizer for about 1 minute. Let most of the solid settle then decant the liquid into the filter. Repeat the operation tow more times. Transfer the filtrate into a 250-ml volumetric flask and dilute to the mark with 90% aqueous methanol. Shake the flask well and pipet 5 ml. of this solution into a 50-ml volumetric flask. Dilute the solution to the mark with 90% aqueous methanol and shake well. Measure the absorbance of this solution in the spectrophotometer at 285nm. Calculate the DPA content of the sample according to the following equation:

$$\%DPA = \frac{(y-a)\ 250}{bw}$$

Where y = absorbance
a = intercept of calibration curve
b = slope of calibration curve
w = weight of specimen, in grams"

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* 4.5.3.3.2.2.1 Delete in its entirety and substitute the following new paragraph:

"4.5.3.3.2.2.1 Calibration procedure. Dry approximately 1 gram of diphenylamine in the microwave moisture analyzer. Weigh approximately 200 mg of dry DPA to the nearest 0.2 mg. Transfer this into a 250-ml volumetric flask. Add 90% aqueous methanol and dissolve completely. Dilute to mark and mix well.

Pipet 5 ml of above solution and transfer into a 100-ml volumetric flask. Dilute to mark with 90% aqueous methanol and mix well. This is the standard stock solution. Pipet volumetrically the following aliquots of the standard stock solution into 100-ml volumetric flasks and dilute each with 90% aqueous methanol to the mark then mix well.

Standard	Standard Stock Solution Aliquot (ml)	Total Volume (ml)	Concentration (mg/ml)
1	5	100	W/100,000
2	10	100	2W/100,000
3	15	100	3W/100,000

Where W = weight of dry DPA, in mg.

Zero the spectrophotometer with fresh 90% aqueous methanol at 285 nm. Rinse the cell with the first standard twice, then measure the absorbance. Repeat the same procedure for standards 2 and 3. Perform a linear regression analysis on absorbance and concentration of standard solutions and obtain the slope, intercept, and correlation coefficient of calibration. If the latter value is less that 0.99 rerun the standards in the spectrophotometer."

* 4.5.3.3.3 Delete in its entirety and substitute the following:

"4.5.3.3.3 Nitrocellulose. Schulze-Tiemann Method for Nitrocellulose (NC) determinations.

Calibrate a eudiometer tube prior to assembly of the Schulze-Tiemann apparatus by suspending the tube in an upside down position and filling this with incremental volumes of aqueous ethanol solution from the buret. Take a reading on the eudiometer tube for each volume of aqueous ethanol transferred, from the buret. Subtract each volume measured with the buret from the corresponding reading on the eudiometer tube. Take the average of the differences between the buret and eudiometer readings. This average is the eudiometer volume correction X.

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The value of X is positive if the buret readings are smaller than the eudiometer readings and vice versa.

Assemble a 250-ml round-bottom flask, gas evolution apparatus, eudiometer tube, NaOH bath, lead band, heating mantle and leveling bulb as shown in Figure I.

Calibrate the apparatus with dry standard A.C.S. grade KNO3. Follow the procedure of analysis below using 0.5-0.7g of standard KNO3 weighed to the nearest 0.1 mg. Calculate the factor of the apparatus according to the following equation.

$$F = \frac{616.8W(273.15+T)}{(V-X)(B-C-P-S)}$$

F= Eudiometer Factor
W= Mass of KNO3 (g)
T= Temperature of condenser water (degrees C)
V= Volume of Gas in Eudiometer Tube (ml)
B= Barometric Pressure (mmHg) Table II
C= Temperature Correction (mmHg) Table II
P= Vapor Pressure of Water over 33% W/W NaOH at T (mmHg) Table III
S= Local Gravity and Sea Level Values Corrected to Standard Values
X= Eudiometer volume correction (ml)

Prepare samples as follows: Cut sample into 1/4 inch pieces or smaller. Dry to constant weight using the microwave moisture analyzer then store in a desiccator.

Fill the eudiometer tube with the NaOH solution by means of the leveling bulb. Weigh between 0.7 and 1.0 gram of specimen to the nearest 0.1 mg and transfer this into the round-bottom flask. Add 30 ml of deionized (DI) water, heat the water to boiling and purge the flask/gas evolution apparatus assembly with steam for 5 to 10 minutes. Remove the heat for a minute then add 25 ml of FeC12 solution, 25 ml concentrated HC1, and 10 ml of DI water.

Place the eudiometer tube over the tip of the gas evolution apparatus and allow the reaction to continue for 20-30 minutes or until the bubbling has stopped. Add 30 ml of DI water through the addition funnel in a similar manner and continue heating for 5 to 10 more minutes. Move the eudiometer tube away from the gas evolution apparatus, turn off the heat, and release the vacuum in the flask by opening the addition funnel stopcock. Allow the eudiometer tube to stand for at least 10 minutes, then read the gas volume on the eudiometer tube to the nearest 0.1 ml at equilibrium pressure with the atmosphere using the leveling bulb.

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Calculate the NC content of sample according to the following equation.

$$\%NC = \frac{2.246F(V-X)(B-C-P-S)}{NW(273.15+T)}$$

V= Volume of Gas in Eudiometer Tube (ml)
X= Eudiometer Volume Correction
B= Barometric pressure (mmHg)
C= Temperature Correction (mmHg) Table II
P= Vapor Pressure of Water over 33% (w/w) NaOH at T (mmHg) Table II
F= S-T Apparatus Correction Factor
N= Percent of Nitrogen in NC (%)
W= Mass of Sample (g)
T= Temperature of Condenser Water (degrees C)
S= Local Gravity & Sea Level Values Corrected to Standard Values

Preparation of FeC12 Solution

Prepare the solution by adding 1000g of FeC12 .4H O, 1200ml DI water, and 70 ml of concentrated HC1 into a 5-liter round-bottom flask. Attach a reflux condenser and reflux for 20 minutes. Cool to room temperature. Slowly add 20 grams of iron filings or powder to the cool solution. When the gas evolution stops, filter quickly. Use A.C.S. grade chemicals.

Preparation of NaOH Solution, 33% (w/w)

Dissolve 330 grams of reagent grade NaOH in 670 grams of DI water.

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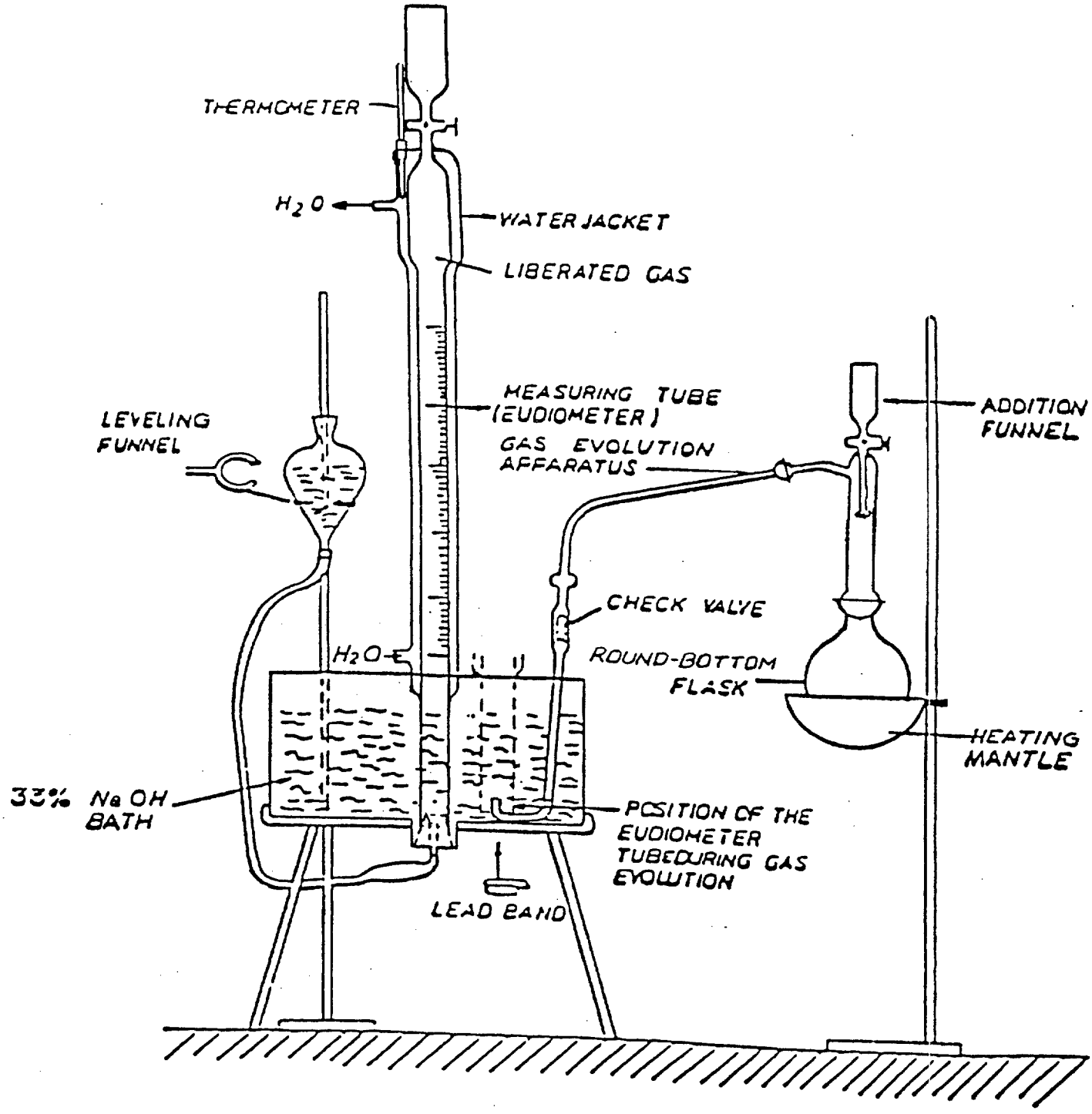


FIGURE 1.

SCHULTZ - TIEMANN - APPARATUS

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TABLE II - BAROMETRIC PRESSURE IN mm Hg

Temp (See Note 1)	740	760	780
14	1.69	1.73	1.78
15	1.81	1.86	1.91
16	1.93	1.98	2.03
17	2.05	2.10	2.16
18	2.17	2.23	2.29
19	2.29	2.35	2.41
20	2.41	2.47	2.54
21	2.53	2.60	2.67
22	2.65	2.72	2.79
23	2.77	2.84	2.92
24	2.89	2.97	3.05
25	3.01	3.09	3.17
26	3.13	3.21	3.30
27	3.25	3.34	3.42
28	3.37	3.46	3.55

Note 1: Temperature (Degrees C) on Barometer Thermometer.

Vapor Pressure of Water Dyer 33% (w/w) NaOH Variable F
Temperature of Condenser Water

(^{o}C)	18	19	20	21	22	23	24	25
P =	11.6	12.4	13.2	14.0	14.9	15.8	16.8	17.8

Barometric pressure temperature correction, Variable C

Table III

Pressure correction value due to the vapor pressure of sodium hydroxide solution in the eudiometer tube.

TEMPERATURE ^{o}C	VAPOR PRESSURE OF NaOH SOLUTION (mm Hg)
18	11.6
19	12.4
20	13.2
21	14.0
22	14.9
23	15.8
24	16.8
25	17.8

Calculation of P_a (actual pressure) given a temperature T_a

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(actual temperature):

$$(T_a - T_1) \times (P_2 - P_1) + P_1 = P_a$$

Where:

T_a = actual condenser water temperature
T_1 = next lowest tabulated temperature value from T_a
P_2 = pressure at next highest tabulated temp from T_a
P_1 = pressure at T_1 "

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* 4.5.3.3.4 Delete in its entirety and substitute the following:

"4.5.3.3.4 Talc content (ash test). The specimen of approximately 1-1.5 grams shall be taken from the same case body for the above testing and weighed to within 0.2 mg in a tared glass or porcelain crucible. Cover the crucible. The specimen shall then be ignited in a microwave furnace (using the glass crucible) at 600 degrees C for 20 minutes or in a muffle furnace (using the porcelain crucible) at 500 degrees C for one hour or until all evidence of carbon is gone (no black residue). The crucible shall be cooled to room temperature and then weighed. The increase in weight of the crucible shall be calculated on a total volatile-free basis to the percent of talc content."

4.5.4 Delete in its entirety and substitute the following new paragraphs:

"4.5.4 Tensile strength

4.5.4.1 Body. Two specimens shall be prepared from each body. The specimens shall be taken from the center of the wall of the body. Test specimens shall be taken 180 degrees apart. The specimens shall have a one-inch gage length and width, but wall thickness shall fall within the tolerance specified on the applicable drawing. The test specimens shall be preconditioned for a minimum of 4 hours at 23 plus or minus (±) 2 degrees centigrade (C) at a relative humidity of 50 ± 5 percent (%). The tensile breaking strength of the specimens shall be determined in accordance with ASTM-D-638, using a cross head speed of one (1) inch per minute. If any specimen ~~shall be rejected~~ applicable requirement, the lot of the body

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4.5.4.2 Igniter container. The igniter container shall be weighed to the nearest 0.1 g. The igniter container shall be cut a 1.5 inch diameter hole in top area and cut several V-slots into the bottom of the igniter container sample. The sample will then be placed in a test fixture that can be mounted in a tensile testing machine with a constant rate of crosshead movement (See Figure A). The tensile breaking strength of the specimen shall be determined by utilizing a crosshead speed of 1-inch per minute. Record the peak load and weight of each specimen. If any specimen fails to meet the applicable requirement, the lot of the igniter container shall be rejected.

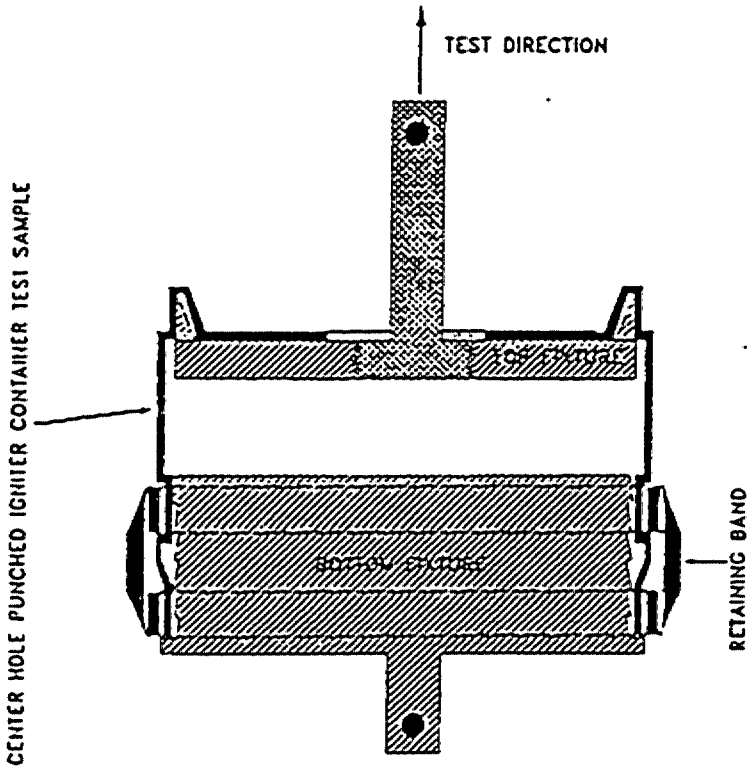


FIGURE A

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4.5.4.3 Rear cap. Two specimens shall be prepared from each rear cap. Test specimens shall be taken from the rear flat of the rear cap. Test specimens shall be taken 180-degree apart. The specimens shall have a 1-inch gage length and width and wall thickness shall fall within the tolerance specified on applicable drawing. Test specimens shall be preconditioned for a minimum of 4 hours at 23 degrees ± 2 degrees C and a relative humidity of 50 ± 5 %. The tensile testing machine shall be of a constant rate of crosshead movement type. Tensile strength shall be determined at a crosshead speed of 1-inch per minute. Record the tensile strength of the specimen. If any specimen fails to meet the applicable requirement, the lot of the rear cap shall be rejected."

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